Algebra 2 Quadratic Functions & Complex Numbers Cheat Sheet

Unit Focus: *parabola forms, completing the square, the quadratic formula, and the discriminant*

1 KEY VOCABULARY

parabola	The U-shaped graph of a quadratic function $y = ax^2 + bx + c$, $a \neq 0$. It is symmetric.
vertex	The turning point (h, k): the **minimum** value if $a > 0$, the **maximum** if $a < 0$.
axis of symmetry	The vertical line $x = h$ through the vertex. It cuts the parabola into mirror halves.
vertex form	$f(x) = a(x - h)^2 + k$. Shows the vertex (h, k) at a glance.
standard form	$f(x) = ax^2 + bx + c$. The y-intercept is c.
factored form	$f(x) = a(x - p)(x - q)$. The zeros are p and q.
zeros (roots)	The x-values where $f(x) = 0$. Real ones are the x-intercepts of the graph.
discriminant	$b^2 - 4ac$. Its sign tells the **number and type** of solutions.
imaginary unit	$i = \sqrt{-1}$, so $i^2 = -1$ (not 1).
complex number	A number $a + bi$: real part a, imaginary part b. Real numbers have $b = 0$.
complex conjugate	$a - bi$ is the conjugate of $a + bi$. Their product $a^2 + b^2$ is real.

2 THREE FORMS OF A QUADRATIC

Vertex	$f(x) = a(x - h)^2 + k$ · vertex (h, k); ***h* has the opposite sign**
Standard	$f(x) = ax^2 + bx + c$ · axis $x = -\frac{b}{2a}$; y-intercept c
Factored	$f(x) = a(x - p)(x - q)$ · zeros p, q; axis $x = \frac{p + q}{2}$

***a* > 0**: opens up (minimum). ***a* < 0**: opens down (maximum).

Example: Rewrite $f(x) = 2x^2 - 12x + 10$

Vertex: $x = -\frac{-12}{2(2)} = 3$; $f(3) = 2(9) - 36 + 10 = -8$, so $2(x - 3)^2 - 8$

Factored: $2(x^2 - 6x + 5) = 2(x - 1)(x - 5)$ zeros 1 and 5; axis $\frac{1 + 5}{2} = 3$ ✓

3 COMPLETING THE SQUARE

add $(\frac{b}{2})^2$ to **both sides**

Example: Solve $x^2 - 6x + 3 = 0$

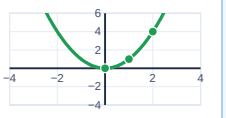
$x^2 - 6x = -3 \rightarrow$ add $(\frac{-6}{2})^2 = 9$ to both sides: $x^2 - 6x + 9 = 6$

$(x - 3)^2 = 6 \rightarrow x - 3 = \pm\sqrt{6} \rightarrow x = 3 \pm \sqrt{6} \approx 5.45, 0.55$

Example: Write $f(x) = 2x^2 + 12x + 7$ in vertex form

Factor out a: $2(x^2 + 6x) + 7$. Add 9 **inside**, which adds $2 \cdot 9 = 18$, so subtract 18: $2(x^2 + 6x + 9) + 7 - 18 = 2(x + 3)^2 - 11$ vertex $(-3, -11)$. If $a \neq 1$ in an equation, divide **every term** by a first.

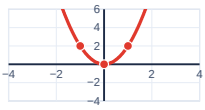
4 GRAPHING PARABOLAS & TRANSFORMATIONS



Parent

$y = x^2$

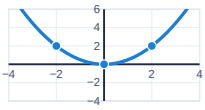
vertex (0, 0), ***a*** = 1



Narrower

$y = 2x^2$

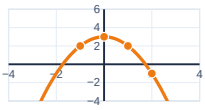
$|a| > 1$: stretched



Wider

$y = \frac{1}{2}x^2$

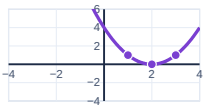
$0 < |a| < 1$: compressed



Reflect & up

$y = -x^2 + 3$

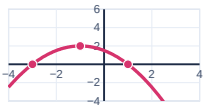
a < 0: down, vertex (0, 3)



Shift right

$y = (x - 2)^2$

vertex (2, 0): **right** 2



Combined

$y = -\frac{1}{2}(x + 1)^2 + 2$

vertex $(-1, 2)$, down, wide

$f(x) = a(x - h)^2 + k$: ***a*** stretches or flips, ***h*** shifts right (+) or left, ***k*** shifts up (+) or down. Dots are lattice points.

5 SOLVING & THE QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example: $3x^2 - 4x - 2 = 0$, $a = 3$, $b = -4$, $c = -2$

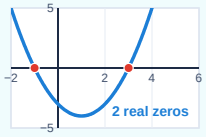
$b^2 - 4ac = (-4)^2 - 4(3)(-2) = 16 + 24 = 40$

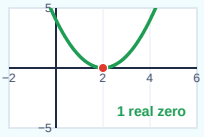
$x = \frac{-(-4) \pm \sqrt{40}}{2(3)} = \frac{4 \pm 2\sqrt{10}}{6} = \frac{2 \pm \sqrt{10}}{3} \approx 1.72, -0.39$

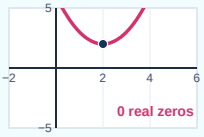
Method	Best when	Example
factoring	it factors	$(x - 4)(x + 3) = 0$: $x = 4, -3$
square roots	no x-term	$2x^2 = 50$: $x = \pm 5$
complete square	$a = 1$, even b	$(x - 3)^2 = 6$

6 THE DISCRIMINANT

$b^2 - 4ac$	Solutions	Example
positive, perfect square	2 rational real	$x^2 - 2x - 3$: 16
positive, not a square	2 irrational real	$x^2 - 2x - 1$: 8
zero	1 real (double)	$x^2 - 4x + 4$: 0
negative	2 complex (conjugates)	$x^2 - 4x + 6$: -8







Zeros of $x^2 - 4x + 6$: $x = 2 \pm i\sqrt{2}$. Negative: no x-intercepts.

7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- In $(x - h)^2$, the vertex x-coordinate is h: $(x + 4)^2$ means $x = -4$.
- Square roots give **two** answers: ±.
- Divide **every** term of the numerator by $2a$.
- Complete the square: add the same amount to **both sides**.

COMMON MISTAKES:

- Writing b^2 as -25 for $b = -5$ (it is $(-5)^2 = 25$).
- Simplifying $\frac{4 + 2\sqrt{10}}{6}$ as $4 + \frac{\sqrt{10}}{3}$.
- Forgetting to factor out a (or to multiply the added number by a).
- Dropping the ± after taking a square root.

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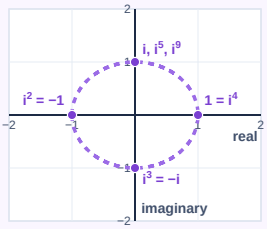
More Practice: *complex numbers, inequalities, linear-quadratic systems, and modeling*

1 IMAGINARY UNIT & POWERS OF *i*

$i = \sqrt{-1}$ $\quad i^2 = -1$

Simplify: $\sqrt{-49} = 7i$ $\quad \sqrt{-18} = i\sqrt{18} = 3i\sqrt{2}$ **(pull out *i* first)**

The cycle repeats every 4:
$i^1 = i$ $\quad i^2 = -1$ $\quad i^3 = -i$ $\quad i^4 = 1$



Example: i^{47}

Divide the exponent by 4 and use the **remainder**: $47 = 4(11) + 3$, so $i^{47} = i^3 = -i$

Remainder 0 means 1: $i^{100} = 1$.
Remainder 2: $i^{58} = -1$. Remainder 1: $i^{21} = i$.

2 COMPLEX NUMBER OPERATIONS

Add and subtract: combine real parts and imaginary parts. **Distribute the minus!**
$(3 + 2i) + (5 - 6i) = \mathbf{8 - 4i}$ $\quad (3 + 2i) - (5 - 6i) = \mathbf{-2 + 8i}$

Multiply (FOIL), then use $i^2 = -1$:
$(3 + 2i)(4 - i) = 12 - 3i + 8i - 2i^2 = 12 + 5i + 2 = 14 + 5i$

Conjugates: $(a + bi)(a - bi) = a^2 + b^2$, e.g. $(3 + 2i)(3 - 2i) = 13$

Example: **Divide:** $\dfrac{8 + i}{2 - i}$

Multiply top and bottom by the **conjugate** $2 + i$: $\dfrac{(8 + i)(2 + i)}{(2 - i)(2 + i)} = \dfrac{15 + 10i}{5} = 3 + 2i$

Absolute value: $|3 + 4i| = \sqrt{3^2 + 4^2} = 5$

3 COMPLEX ROOTS & WRITING QUADRATICS

Example: **Solve $x^2 - 4x + 13 = 0$**

$b^2 - 4ac = 16 - 52 = -36 < 0 \rightarrow$ complex roots

$x = \dfrac{4 \pm \sqrt{-36}}{2} = \dfrac{4 \pm 6i}{2} = 2 \pm 3i$ **(divide both terms)**

Real-coefficient roots come as **conjugate pairs**. Check: sum $4 = -\dfrac{b}{a}$, product $(2 + 3i)(2 - 3i) = 13 = \dfrac{c}{a}$.

From the roots: $a(x - p)(x - q)$.
Roots $2 \pm 3i$ give $(x - 2 - 3i)(x - 2 + 3i) = x^2 - 4x + 13$.

From the vertex and a point: vertex (3, −8) through (1, 0)
$0 = a(1 - 3)^2 - 8 \rightarrow 4a = 8$, $a = 2 \rightarrow$ $\boldsymbol{f(x) = 2(x - 3)^2 - 8}$

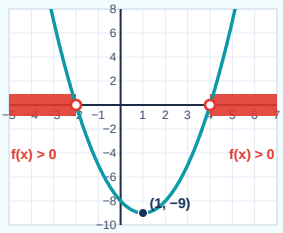
4 QUADRATIC INEQUALITIES

Example: **$x^2 - 2x - 8 > 0$**

1. Solve $x^2 - 2x - 8 = 0$: $(x - 4)(x + 2) = 0$, so $x = -2, 4$
2. Sketch: opens up, so $f(x) > 0$ **outside** the zeros
3. Test a point from each region: $-3 \rightarrow 7$, $0 \rightarrow -8$, $5 \rightarrow 7$

> 0: $(-\infty, -2) \cup (4, \infty)$ $\quad \leq 0$: $[-2, 4]$

Open circle and parenthesis for $< >$; **closed** circle and bracket for $\leq \geq$. If it opens **down**, the regions swap.



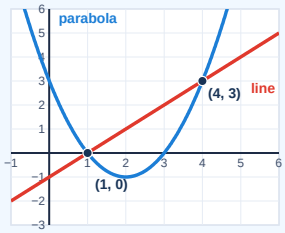
$x^2 + 1 < 0$ has no solution; $x^2 + 1 > 0$ is true for all reals.

5 LINEAR-QUADRATIC SYSTEMS

Example: **$y = x^2 - 4x + 3$ and $y = x - 1$**

Substitute: $x^2 - 4x + 3 = x - 1 \rightarrow x^2 - 5x + 4 = 0 \rightarrow (x - 1)(x - 4) = 0$
Put each x in the **line**: $y = 0$ and $y = 3$.
(1, 0) and (4, 3)

Combined quadratic	Meaning
$b^2 - 4ac > 0$	2 points
$= 0$ (e.g. $y = x^2$, $y = 2x - 1$)	1 point (tangent): (1, 1)
< 0 (e.g. $y = x^2$, $y = 2x - 2$)	0 real points



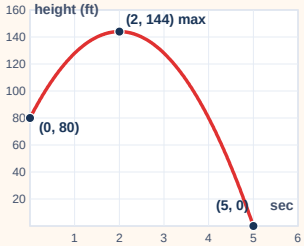
6 QUADRATIC MODELING

Example: **A ball leaves a roof: $h(t) = -16t^2 + 64t + 80$ (h in feet, t in seconds)**

Starting height: $h(0) = 80$ ft. **Max:** $t = -\dfrac{64}{2(-16)} = 2$ s, $h(2) =$ **144 ft** (the max is the h-value, not 2).

Lands: $-16(t - 5)(t + 1) = 0 \rightarrow t = 5$ or -1. 5 seconds Reject $t = -1$: time cannot be negative.

Fence: 100 ft of fence, width x: $A = x(50 - x)$, vertex at $x = 25$, so a 25 ft by 25 ft pen has the most area: **625 ft²**.



7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:
- $i^2 = -1$. Reduce a power of i with the remainder after dividing by 4.
- To divide, multiply by the **conjugate** of the denominator.
- Answer a modeling question with the right quantity and units (max **value**, time, ft).
- Test a point in each region of an inequality.
- Check solutions of a system in **both** equations.

COMMON MISTAKES:
- Treating i^2 as 1, or writing the conjugate with the wrong sign.
- Dividing complex numbers without a conjugate.
- Choosing the wrong region (or the wrong end-point type) for an inequality.
- Reporting the vertex x as the maximum value.
- Accepting $t < 0$ as a real time, or rejecting a negative root that makes sense.